



University of
BRISTOL School of Physics

LEVEL 6 MSci LABORATORY

HANDBOOK

2024/2025



Introduction

The MSci Laboratory is a part of the PHYS30033 Practical Physics III: Research Skills and Group Project 2024 unit and provides a range of experiments to develop your experimental research and report writing skills. The experiments are more demanding than those in Year 2 and in many cases allow students to use advanced instrumentation and research methods. Experiments are generally associated with particular research groups or topic, with an academic supervisor from the research group responsible for each experiment.

All information about the course is given on the Blackboard under:

[Practical Physics III: Research Skills and Group Project 2024](#)

Timetable

You will be required to carry out **one experiment**. There are 6 in-lab sessions in total for each experiment (experiment weeks 1-6). Each experiment contains also online material on the Blackboard for study: pre-lab online introduction* (week 0/1, before the experiment starts) and additional online material for study during weeks 1-6. There are 3 lab groups: **Group A (Thursdays)** or **Group B (Fridays)**, timetable weeks 1-6 in TB1 and **Group C (Thursdays)** weeks 13-18 in TB2.

Laboratory hours are 10:00-13:00 and 14:00-17:00.

* - The online pre-lab session contains material which you must engage with before the start of the lab. This session is not timetabled but might contain a synchronous online session with your supervisor. In this case he/she will inform you about the time and invite you to it.

Attendance

If you are physically present in the lab for any of the sessions you must sign-in at the beginning of each morning and afternoon session and sign-out, accordingly, using QR code or URL provided. If your lab session is online, you shall use the weekly link to the sign-in form and sign as "working online". Satisfactory attendance is required to pass the Unit. In addition, **your electronic notebook must reflect the work done in each week** and not be filled retrospectively as an afterthought.

Selection of Experiment

This needs to be completed by **12:00 Monday, 16 September Week 1**.

- Listen to the introductory presentation online on Blackboard: MSci Laboratory section.
- Find in which Lab group you are (A,B,C). You have been already allocated a lab day by the central team (e.g. Friday, Group B). Some of you have given us your preference for a lab partner. If not, you might want to contact someone in your group and see if they are happy to be in a pair with you. If no lab partner found - don't worry, just follow other steps below.
- Read the description of all experiments in this Handbook. Discuss with your lab partner, if you have one.
- Submit your [preferences online](#). You should sign up, giving your preferred laboratory partner, for either group A, B or C. Only one person per pair shall submit this form. If you have no partner, just input NONE and we will pair you up. On the online form you should indicate your preferred experiments in the rank

order. As there are only a fixed number of apparatus sets, we may not be able to allocate your first preferred choice. Allocation of experiments will be finalised after the sign-up process is completed. In the event of an experiment being over-subscribed, the allocation will be performed randomly. Normally, students on the Physics with Astrophysics programme will be given preference for the Astrophysics experiments, although all students can put them in their list of preferences. Note, Theory experiments are only for Theoretical Physics MSci programme.

- Your preferences must be indicated **by 12:00 on Monday, 16 September 2024 in week 1.**

Notebook

Each individual student is responsible for keeping **their own** laboratory notebook. Each of you must submit **an electronic labbook** for this unit.

A good notebook should be an experiment diary, documenting what you did on every day, *in and outside the lab, written down as you went along*. It is important for your notebook to be comprehensible to someone other than yourself, but some “untidiness” is expected. If you are working as a pair, your notebook should be maintained separately from your partner as much as it practically possible. It is your personal record of the experiment and will be used for assessment of your work in the lab.

The notebook must contain clear record of what you have been doing on a daily basis, with notes on aims, theory, results etc. Even for a computational project there is still scope for making written notes. The degree to which the code is commented on, and sensibly organised, will also be considered. For further guidance, please see the assessment notes in the appendix of this handbook.

Your supervisor’s assessment of your work in the lab will be based on your organisation and logical approach to the experiment; your lab notebook is a primary indicator of this.

Electronic lab books

The format of MSci lab electronic labbook is based on the OneNote Microsoft M365 package. This is very versatile tool allowing integration of notes, sketches, graphs, links, images and videos, as well as research data in one package. The guidelines for a good electronic lab-book keeping are similar to those for paper notebooks. It is advisable, that a **good and clear structure followed inside the lab book file**, e.g. separate page notes for each lab day, experimental methods, analysis, theory, extra material, etc. Please, get yourself familiar with the Office 365 package before the lab start. There are different versions of OneNote (see Blackboard), make sure you use the appropriate one. All students will be required to produce an electronic lab book at the end of the lab for assessment.

N.B. Even if you used paper-based notes throughout the lab, you must scan or take good quality photos of these notes and insert them into the OneNote document as images without reduction in size. If you have a paper lab book in addition to the electronic version: **Do not work on loose sheets of paper or in a rough book (you will be penalised if you do).** Your additional notebook should be a bound book so that sheets of paper cannot be added or removed. You should not have recorded data on scraps of paper – or in the back of your notebook, even if it is written up later.

You shall submit this OneNote document on Blackboard unit as a **single pdf file**. You shall also share the original via Microsoft tools (e.g. OneDrive) with you supervisor in the week 1 (please allow editing so supervisor can insert comments). **Each student should submit their own electronic lab book**. Though it is possible to share some notes on the experiments and data, all analysis and conclusions should be individual work. Please make sure you **reference any work/material which is not your own**, such as texts, images, data, graphs or external resources you use in the electronic lab book appropriately.

Assessment

Your assessment will be based on following evidence: your **notebook** containing your laboratory notes written during work on the experiment and analysis during practical work period, a **report** written after the experiment, as well as a **day-to-day performance** in the laboratory. The assessment will utilise following criteria:

- **Knowledge & Understanding:**

Content knowledge; Critical approach. Assesses level of knowledge and mastery of the relevant physics, including backgrounds of theory, experimental and analytic methods, ability to link them with wider discipline areas and applications.

- **Intellectual Skills:**

Use of evidence & sources; Explanation, application & interpretation; Problem-solving.

Use of evidence & sources: Ability to thoroughly evaluate, critically engage with, and effectively utilise a wide range of relevant sources: own findings, scientific research literature, textbooks, previous learning, etc.

Explanation, application & interpretation: Ability to combine own findings, concepts and theories to procedures, interpretations and analysis. Ability to deploy logical and scientifically rigours arguments to draw conclusions. Ability to focus on relevant and significant aspects of the topic/discussion. Ability to formulate logical and scientifically informative conclusions and suggestions for future work.

Problem-solving: Ability to identifies key aspects of problems. Ability to show initiative, independence and creativity in applying relevant and effective approaches to address scientific challenges.

- **Research & Scientific Practices:**

(Specialist skills & techniques; Research; Academic skills).

Specialist skills & techniques: Mastery of use of specific and relevant research skills and techniques in the work.

Research: Grasping, framing and creating research aims and scientific motivation for the work. Use and extent of application of appropriate preparation and training, experimental and computational methods to obtain the results under supervision. Quality and extent of results obtained and their description. Assesses strength and depth of the analysis and interpretation performed, scientific quality of discussion, cohesion between obtained results, theory and discussion, scientific robustness and quality of conclusions, ability to put conclusions in the context of research aims as well as wider implications.

Academic skills: Use of academic integrity and conventions including proper citation, referencing, and adherence to scholarly standards. Communication in the style appropriate for a given context and audience.

- **Professional and Life Skills.**

Teamwork/Collaboration: Ability to collaborate with others, e.g. students, supervisors, technical specialists, etc., to effectively perform tasks and achieve goals. Assesses individual contribution, leadership and creativity when working with others. Responsiveness and ethical awareness of other's needs and interests. Ability to manage and mitigate conflicts.

Communication: Ability and creativity to communicate aims, background, research methods and results of the work in a clear, cohesive and professional style and manner, using a wide range of appropriate tools: writing, oral presentations, digital technologies and IT, etc.

Insight: Assess ability to work independently and set own goals, critically assess own strengths and weaknesses. Able to identify and implement an effective programme of self-development and training to improve practical and professional skills.

Self-management: Ability to perform all tasks professionally and on time. Self-organisational skills.

You will be given a written feedback, which addresses the criteria above and a single overall mark on the 21 point scale.

The report should be in the style of a scientific research letter paper in the American Physical Society's *Physical Review*. The report should explain what you did and be comprehensible to a *physicist* in another field. Hence you should introduce the experiment appropriately. Some notes on the style and content of the report are given below, however the best way to learn the style in which professional scientific papers are written is to read several actual published papers.

It is strongly recommended that you use the *Latex* document preparation system to produce your report. You should use the RevTex 4 package to get the report in the correct format. The required style is a two-column format, using a 10pt font, and should not be longer than **4 pages of A4**. More details of the required format and a template can be found on Blackboard. **The 4-page maximum includes all graphs, figures, references and appendices** (not recommended for 4 page article). Exceeding this page limit will be considered poor style and your mark will be reduced accordingly.

The report should be submitted electronically via Blackboard for assessment. There is no need to submit a separate paper copy. However, please ensure that the report prints out correctly and does not require special system resources, fonts, etc. As the notebook forms also a part of the assessment, it should also be submitted electronically before the deadline.

The report must be written and prepared individually. Report is not a collaborative project with your partner. This includes all diagrams and figures. You are advised that we use anti-plagiarism software to scan all submitted work. This compares your work to others that have been submitted in your group or previously by other students and also to the resources on internet. You are reminded that plagiarism is taken very seriously by the University of Bristol and usually results in a minimum penalty of a mark of zero been awarded for the unit.

For late submissions standard University penalties will be applied (-10 marks/100 per working day). For work submitted seven calendar days after the submission deadline the student will

receive a mark of zero. Late work will still need to be completed and handed in, in order to satisfy the requirements for passing the unit.

If you believe your mark is unfair, then you have the right to ask for it to be reviewed by the Laboratory co-ordinator (or his delegate). Please note, that as a consequence of this review your mark could be revised up or down.

Deadlines:

All deadlines are at 12.30pm on the day indicated:

Groups A/B – **Monday 11th November** (week 9)

Group C – **Monday 10st March** (week 21)

Laboratory Staff

Position	Name		E-mail address
		Room	
Laboratory Coordinator	Dr Andrei Sarua	3.39	a.sarua@bristol.ac.uk
Technical Support	Tom Kennedy	2.18	Tom.Kennedy@bristol.ac.uk

The Experiments

Code	Title of Experiment	Supervisor	Pairs
AFM	(AFM) Atomic Force Microscopy	Dr Massimo Antognozzi	1
CR	(CR) Properties of Cosmic Ray Muons	Dr Konstantinos Petridis	1
FIS	(FIS) Fourier Image Spectroscopy of Microstructures	Dr Ian Lindsay	2
GC	(GC) Galaxy Clusters	Prof Malcolm Bremer	2
JN	(JN) Johnson Noise	Prof Neil Fox	2
MAP	(MAP) Characterization of Monolithic Active Pixel Sensors	Prof Jaap Velthuis	1
NMR	(NMR) Pulsed Nuclear Magnetic Resonance	Dr Adrian Barnes	2
OVS	(OVS) Optical Vibrational Spectroscopy	Dr Andrei Sarua	1
QTS	(QTS) Quantum Tunnelling in Superconductors	Prof Sven Friedemann	1
SEM	(SEM) Secondary Electron Microscopy	Dr Matt Smith	1
STM	(STM) Scanning Tunneling Microscopy	Dr Shuqiu Wang	2
TE* Group C TB2only	(TE) TESS exoplanet time series data analysis	TBC	2
IM	Theory Labs (Theoretical Physics only) (IM) Ising Model	Dr Francesco Turci	2
LB	(LB) Lattice Boltzmann Method	Dr Alan Raynolds	2
WT	(WT) The Wetting Transition	Dr Stephen Clark	2

Group A: Thursday, weeks 1-6

Group B: Friday, weeks 1-6

Group C: Thursday, weeks 13-18 (Theoretical Physics MSci, Physics with Astro MSci, Physics with Scientific Computing MSci)

N.B. (TE) TESS exoplanet time series data analysis – is only available in Group C.

(AFM) Atomic Force Microscopy

Dr Massimo Antognozzi



This experiment will make use of an Atomic Force Microscope (AFM). The goal is to familiarize the student with the basic principles and operation of this instrument. Different specimens will be investigated, from mineral surfaces to biological molecules (DNA and proteins). The understanding of the mechanical behaviour of micro-scale force sensors is crucial to correctly interpret the AFM results, for this reason part of the project will be dedicated to the development of techniques to investigate the mechanical response of AFM cantilevers. You will also learn some programming in the LabVIEW environment.

(SEM) Secondary Electron Microscopy

Dr Matt Smith

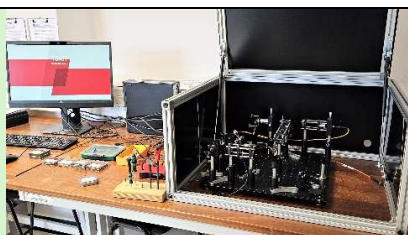


Scanning Electron Microscopy (SEM) uses a beam of high energy electrons to provide an image of the surface of a material at a higher resolution than optical microscopy. In addition, the interaction of the electron beam with the surface releases X-rays which are characteristic of the elemental identity of the surface, and back scattered electrons which can give information about the orientation of the crystal structure.

The teaching lab has installed a new TESCAN SEM and this experiment will use this instrument to analyse a range of different materials. Students will learn how to use the secondary and backscattered electrons as well as emitted X-rays to investigate the microstructure of samples including insects, a gas turbine blade, chemical vapour deposited diamond and a GaN-based transistor. They will explore the effect of the beam voltage and current on the resolution of their images, and how the beam voltage can affect the interaction volume of the electrons with the sample and hence its sensitivity to elemental composition with different depths. Finally, the students will have the chance to choose one of the samples to explore in detail, using the acquired SEM skills and supporting scientific literature to investigate and report on the relevant physics active in the sample of choice.

(FIS) Fourier Image Spectroscopy of Microstructures

Dr Ian Lindsay



Imaging “small” structures is of great interest for many fields of science. Optical microscopy allows one to study structures far smaller than observable with the naked eye; however, the resolution limit of optical microscopes is typically limited to the wavelength of the light being used. To image smaller structures it is often necessary to use expensive and complex techniques such as electron microscopy.

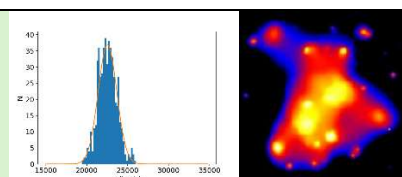
However, information about some sub-wavelength structures may be gained if they are periodic. For example, a diffraction grating may have groove spacing smaller than the wavelength of light, but the angle at which a beam of light is diffracted will give information about the groove spacing. In this experiment you will use a technique called “Fourier image microscopy” to study the angular dependence of the reflectivity of a variety of structures, revealing that the optical

properties of many surfaces are governed by periodic (or almost periodic) sub-wavelength structures.

You will first set up the Fourier imaging experiment by using a grating of a known pitch to calibrate your system. This will give you a practical insight into how lenses perform Fourier transforms, as well as giving you practice as aligning optical systems. You will then study the reflectivity of a variety of structures. These may include artificial structures such as CDs and DVDs which are effectively gratings, and natural samples. These may include opals, which consist of sub-micron sized tightly packed spheres which show a wavelength-dependent angular reflectivity, and biological iridescent structures such as butterfly and beetle wings, whose colour comes not from dyes or chemical colour but from sub-wavelength periodic structures on their wings. This experiment is based on a very similar one used in the Photonics group to study reflectivity of man-made photonic crystals and biological iridescent structures. You will use very similar techniques, studying structures that are still not yet well-understood by the research community. The project will give you an insight into how biology uses structure to control light, as well as the man-made structures we design for photonic circuits and light control for lasers and solar cells.

(GC) Galaxy Clusters

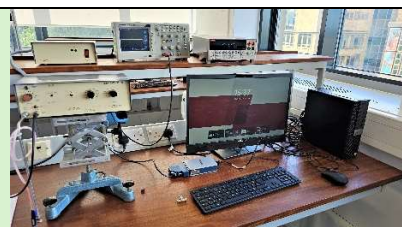
Prof Malcolm Bremer



Clusters of Galaxies are the largest gravitationally bound objects in the Universe and are unique laboratories with which to study plasma physics, galaxy evolution and cosmology. In this experiment you will use measurements of the velocities of galaxies in a cluster to estimate the total mass of the cluster. Combined with other measurements of the amount of baryonic matter in the cluster, you will then estimate the matter density of the Universe. This experiment is based on the analysis of data from online databases and can be performed with basic software tools but is a good opportunity to practice using Python to perform astronomical data analysis. This experiment will also show you how to use simple numerical methods to perform error analysis, which are very powerful and widely applicable.

(JN) Johnson Noise

Prof Neil Fox



Quantum mechanics dictates that electrons inside conductors are continually scattering between states with different energy and momentum. The size of the spread in energy accessible to different electrons is controlled by the Pauli Exclusion Principle and the temperature. These quantum thermal fluctuations actually manifest themselves in a remarkably simple fashion, namely they produce fluctuations in the current and thus the voltage measured across the conductor. These voltage fluctuations are called Johnson noise. The amplitude of the fluctuations is uniquely determined by Boltzmann's constant the resistance of the conductor and the temperature. Hence, this effect can be used to produce an *absolute* thermometer (i.e. one that relies only on the values of fundamental constants and does not require calibration against a standard). Such absolute thermometry effects are actually quite rare and so Johnson noise occupies a key position in Metrology - the science of measurement.

In this experiment you will study these voltage fluctuations, using spectral analysis techniques (Fourier transforms), and determine a value for Boltzmann's constant. You can then use the effect to produce an absolute temperature measurement. This will then be tested by changing the temperature of the resistor, either by heating or cooling with liquid nitrogen. A further aim of the experiment is to give a practical introduction to interfacing data acquisition electronics to a

computer, which may be of great use to you in the future (for example, in your final year project). You will gain experience in the practical use of Fourier transforms as a tool to analyse complex spectra. This project requires computer programming in the language Python.

(MAP) Characterization of Monolithic Active Pixel Sensors

Prof Jaap Velthuis



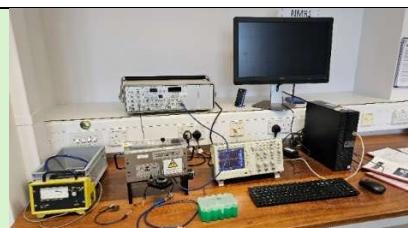
Monolithic Active Pixel Sensors (MAPS) are a novel and exciting development in imaging sensors. They are used for example in the camera of your mobile phone. They are low power, can be made very thin ($<20\mu\text{m}$) with still very good performance and it is possible to incorporate signal processing on the chip. Dedicated sensors are developed for **Particle Physics** mainly with the Linear Collider in mind. For this application it is of paramount importance that the sensor will detect all traversing particles and that their hit position can be reconstructed with excellent precision.

In this lab experiment you will be given a set up with a MAPS device. You will learn about various characterization techniques using light and radio-active sources.

You will study important variable like the noise, the signal, the thickness of the sensitive layer and the size of the charge cloud in the device. You will need to develop your own data analysis software in Python, C or C++ and ROOT. After learning the characterization techniques, you will focus on one aspect and study this in detail.

(NMR) Pulsed Nuclear Magnetic Resonance

Dr Adrian Barnes



Nuclear magnetic resonance is a powerful probe used in physics, chemistry and biological research. The applications are widespread, ranging from the identification and structure determination of complex molecules, imaging of the human body, and the study of magnetic fluctuations in metals. This experiment will provide an introduction to the NMR technique using a commercial teaching spectrometer. Various different techniques will be used to measure the NMR relaxation rates of protons in water. The experiment will then go on to apply these techniques to investigating how the resonance changes with the environment of the proton nucleus.

(OVS) Optical Vibrational Spectroscopy

Dr Andrei Sarua



This experiment studies interaction between photons of light and quantised vibrations of atoms in matter, i.e., phonons. When a laser beam is incident on a material the light can scatter either elastically (the scattered light is at the same frequency as incident beam) or inelastically (with a change in frequency). The latter is known as Raman effect and the frequency shift of scattered light depends on the interactions of incident photons with electrons and vibrational excitations in the material. This is enabling access to the information about chemical bonds, structure,

composition, stress, temperature, etc., in the materials. Hence, Raman spectroscopy is an extremely versatile, non-destructive and widely used analytical tool in the material science, chemistry as well as in industry: e.g., in forensics, environmental monitoring and process control. Since 2020 there is even a Raman spectrometer on Mars (SHERLOC instrument on NASA Perseverance rover).

In this experiment, you will examine the phonons and phonon dynamics in different solid crystalline materials by recording their Raman spectra. Spectra will be acquired using a PerkinElmer Raman spectrometer, similar to the equipment you can find in many research and analytical labs. Here the laser beam is focused on the sample surface using optical microscope into a micrometre sized spot, which allows studying the matter on a small scale. This is a multiphysics project: covering quantum and macroscopic theory of solids, light-matter interactions and practical aspects of laser spectroscopy, optics and instrumentation.

(QTS) Quantum Tunnelling in Superconductors

Prof Sven Friedemann

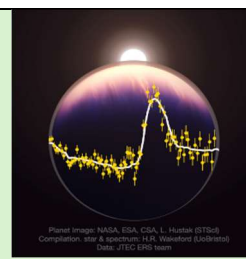


A surprising result of the theory of superconductivity is that when two superconductors are placed in close proximity, separated by a thin insulating barrier, a supercurrent can flow between the superconductors despite the presence of the insulating barrier. The occurrence and strange behaviour of this current as a function of applied dc voltage, and magnetic field, is known as the Josephson effect. The effect is entirely quantum mechanical in nature and is used as a basis of many important applications of superconducting devices, such as the most precise magnetometers currently known and devices which offer the most precise method of measuring the ratio of fundamental constants, h/e . A related effect is single electron (Giaever) tunnelling, which dominates when the insulating barrier is thicker. The aim of this experiment is to investigate these effects in superconductors at liquid helium temperatures. Samples will be fabricated using thin film evaporation techniques.

(TE) TESS exoplanet time series data analysis

TBC

Group C, TB2 only



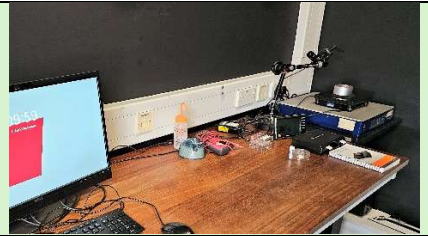
Exoplanets, planets that orbit other stars, have been discovered in their thousands in just the past two decades. Currently the *Transiting Exoplanet Survey Satellite* (TESS) is in orbit around the Earth surveying the whole sky for exoplanets that periodically transit their bright host stars. TESS has imaged many thousands of stars producing high cadence time series data in ~month long sectors of the sky. In this experiment you will examine a series of TESS time series light curves to search for signatures of astrophysical phenomena such as eclipsing binary stars, stellar activity (flares, rotation, variability), and transiting exoplanets. You will approximate and measure the periodicity of signals in the data to determine their astrophysical origin to measure the target star and system characteristics. Using this information along with fundamental laws of planetary motion and physics we can determine the period, orbital distance, and size of discovered transiting exoplanets (or stars) in the data.

Key learning objectives include: getting hands of experience with space-based time series data; understand and identify by eye any astrophysical phenomena apparent in a range of datasets; be

able to examine in detail periodic repeating phenomena extracting precise periods and amplitude changes in stellar flux detailing their origins.
This experiment will require moderate/good python coding skills for the whole project making use of the *Python lightkurve* software package.

(STM) Scanning Tunneling Microscopy

Dr Shuqiu Wang

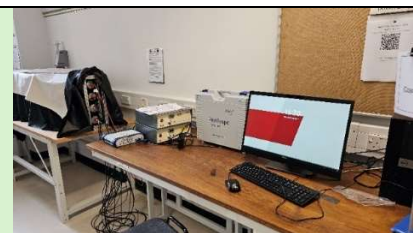


In 1982 a paper was published announcing the successful measurement of the tunnel current between a tungsten and a platinum electrode in adjustable vacuum conditions. This result was the first step towards scanning tunnelling microscopy (STM), which was officially born a few months later with the publication of a first image of a metallic surface. The next paper showed, for the first time, atoms imaged in real space using STM. In 1986 Binnig and Rohrer were awarded the Nobel prize in Physics for developing this technique. STM is a very good example of how a fundamental understanding of quantum mechanics can be used to produce significant technological innovation.

In this experiment you will use a STM to image the surface of crystals with atomic resolution. This will teach you the principles of the STM technique as well as how it is used in a practical situation. In the second part of the project, you will use STM to probe the electronic properties of technologically important materials.

(CR) Properties of Cosmic Ray Muons

Dr Konstantinos Petridis

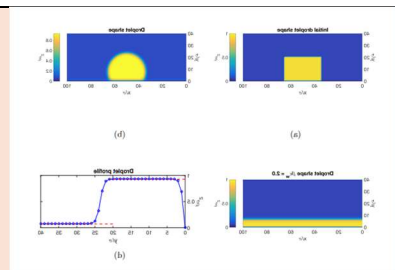


Whilst our lab is too small to accommodate an accelerator-based particle physics experiment, we can make use of another handy source of high-energy particles: cosmic rays. A stream of charged particles, mostly muons, continually passes through us at ground level. By stopping some of these muons in a heavy absorber and employing scintillation detectors, we can observe their decay and measure their lifetime. This project is a particle physics experiment "in miniature", and uses the type of apparatus that was winning Nobel Prizes only fifty years ago. It provides a good introduction to the detector techniques used in modern particle, nuclear and astrophysics, and to the type of statistical analysis used in these areas. The experiment will develop experimental skills in electronics hardware, computing (Python) and in data analysis.

Further details and experimental resources/ scripts for some of the experiments can be found on Blackboard: <https://www.ole.bris.ac.uk/>

(WT) The Wetting Transition

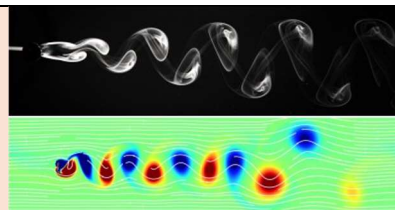
Prof Stephen Clark



The behaviour of liquids at interfaces and in confinement is a fascinating and important area of study. For example, the behaviour of a liquid under confinement between two surfaces determines how good a lubricant that liquid is. The nature of the interactions between the liquid and the surfaces is crucial. Consider, for example, the Teflon coating on non-stick cooking pans, used because water does not adhere to "wet" the surface. In this theory experiment we will develop a discrete mean-field approximation for the behaviour of a fluid called density functional theory. The aim of this project is two-fold: (i) to derive DFT for an inhomogeneous lattice-gas fluid, whilst explaining the physics of the theory; (ii) To illustrate the types of quantities that DFT can be used to calculate, such as the surface tension of the liquid-gas interface, to study wetting behaviour and to answer the question "what is the shape of a drop of liquid on a surface?".

(LB) The Lattice Boltzmann Method

Dr Alan Reynolds

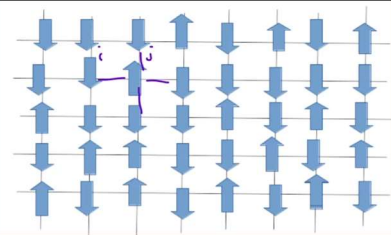


Accurate modelling of the fluid dynamics can be challenging even in simple setups, such as flow past a fixed, rigid obstacle. Conventional, macroscopic approaches attempt to numerically solve the Navier-Stokes equations, yet the non-linearity of the equations, complex geometries and the intricacies of turbulent flow can cause difficulties. A microscopic approach, tracking the position and velocity of individual molecules, clearly results in far too many degrees of freedom to feasibly handle. The lattice Boltzmann method is a compromise between the macroscopic and microscopic views. Space is discretised into a lattice, with the fluid at the lattice points described by a simple kinetic model. It is surprisingly easy to code, yet can reproduce complex fluid dynamics such as vortex shedding.

In this project, you will be guided through much of the implementation of the lattice Boltzmann method. You will then be encouraged to adjust and adapt the resulting code to enable an exploration of the features of fluid flow past a selection of obstacles.

(IM) Ising Model

Dr Francesco Turci



The Ising model represents a simplified magnet in which each atom has a spin, which can only take one of two values, 'up' and 'down'. Historically, it played a fundamental role in the theory of phase transitions in statistical physics, since it goes from a non-magnetic state at high temperature to a magnetic state at low temperatures, below the Curie temperature T_c . But there are many surprises, and the model and its close relatives are still central to modern physics. For example, the 2016 Nobel prize was awarded to Kosterlitz and Thouless for the theory of phase transitions in the related XY model (where the spin is still fixed in size, but can have any direction in the x-y plane). The Ising model is discrete and so is particularly suited for investigation by computers. In this laboratory we shall investigate the thermodynamics of the Ising model using the Metropolis Monte-Carlo method. Using this approach, we shall investigate the Ising model in zero, one and two dimensions, seeing how the statistical state changes as temperature and magnetic field are varied. In particular, we will use the Metropolis method to investigate the 'critical exponents' in the two-dimensional model, which represent how the magnetization, energy, entropy, heat capacity, etc., change near to the Curie temperature T_c . Optionally, the role of the lattice geometry, dimensionality, and similar models, such as Potts and XY could also be explored. This laboratory is specific to theoretical physics students, and introduces concepts, which will be fundamental in the later, e.g., 4th year Phase Transitions course.

Further details and experimental resources/ scripts for some of the experiments can be found on Blackboard: <https://www.ole.bris.ac.uk/>

PHYS30030 Practical Physics III: Research Skills and Group Project
MSci Practical Laboratory

Student:
Experiment:
Assessor/Supervisor:
Date:

Mark (0-20):

This mark is provisional and may be subject to moderation. Marking criteria are given in the unit Handbook.

Feedback

The letter grades below summarise the level attained in each category:
A*=20-18, A=15-17, B=12-14, C=9-11, D=7-8, E=1-6, 0=0

Knowledge & Understanding (A*-E):
(Content knowledge; Critical approach)

Intellectual Skills (A*-E):
(Use of evidence & sources; Explanation, application & interpretation; Problem-solving)

Research & Scientific Practices (A*-E):
(Specialist skills & techniques; Research; Academic skills)

Professional & Life Skills (A*-E):
(Teamwork; Communication; Insight; Self-management)